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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 11.4

$$f(x) = \frac{3x^2 - 12x + 9}{x - 2}$$

1. Domein van $f(x)$:

$$\text{dom}(f) = \mathbb{R} \setminus \{2\}$$

2. Asymptoten:

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{-3}{0} = +\infty$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{-3}{0} = -\infty$$

Er is dus een verticale asymptoot in $x = 2$. Langs links gaat f naar $+\infty$,

langs rechts naar $-\infty$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{3x^2 + 12x + 9}{x^2 - 2x} = 3 = a$$

$$\lim_{x \rightarrow \infty} f(x) - ax = \lim_{x \rightarrow \infty} \frac{3x^2 - 12x + 9 - 3x^2 + 6x}{x - 2} = \lim_{x \rightarrow \infty} \frac{-6x}{x} = -6$$

$$\text{SA: } y = 3x - 6$$

geen HA

$$f(x) - 3x + 6 = \frac{\cancel{3x^2} - \cancel{12x} + 9 - \cancel{3x^2} + \cancel{6x} + \cancel{6x} - 12}{x - 2} = \frac{-3}{x - 2}$$

TO:

x	2		
T	-	-	-
N	-	0	+
$f(x) - 3x + 6$	+		-

dus f ligt boven SA bij $-\infty$ en onder SA bij $+\infty$

3. Tekenonderzoek van $f(x)$:

$$D = 144 - 4 \cdot 3 \cdot 9 = 36$$

$$x_1 = \frac{12 + 6}{6} = 3$$

$$x_2 = \frac{12 - 6}{6} = 1$$

TO:

x	1			2			3		
T	+	0	-	-	-	0	+		
N	-	-	-	0	+	+	+		
$f(x) - 3x + 6$	-	0	+		-	0	+		

4. Tekenonderzoek $f'(x)$:

$$f'(x) = d\left(\frac{3x^2 - 12x + 9}{x - 2}\right) = \frac{(6x - 12)(x - 2) - (3x^2 - 12x + 9)}{(x - 2)^2}$$

$$= \frac{6x^2 - 12x - 12x + 24 - 3x^2 + 12 - 9}{(x - 2)^2} = \frac{3x^2 - 12x + 15}{(x - 2)^2}$$

T: $D = 144 - 4 \cdot 3 \cdot 15 = -36$

x	2		
T	-	-	-
N	-	0	+
$f'(x)$	+	⁽²⁾	-
$f(x)$	↗		↗

5. Tekenonderzoek $f''(x)$:

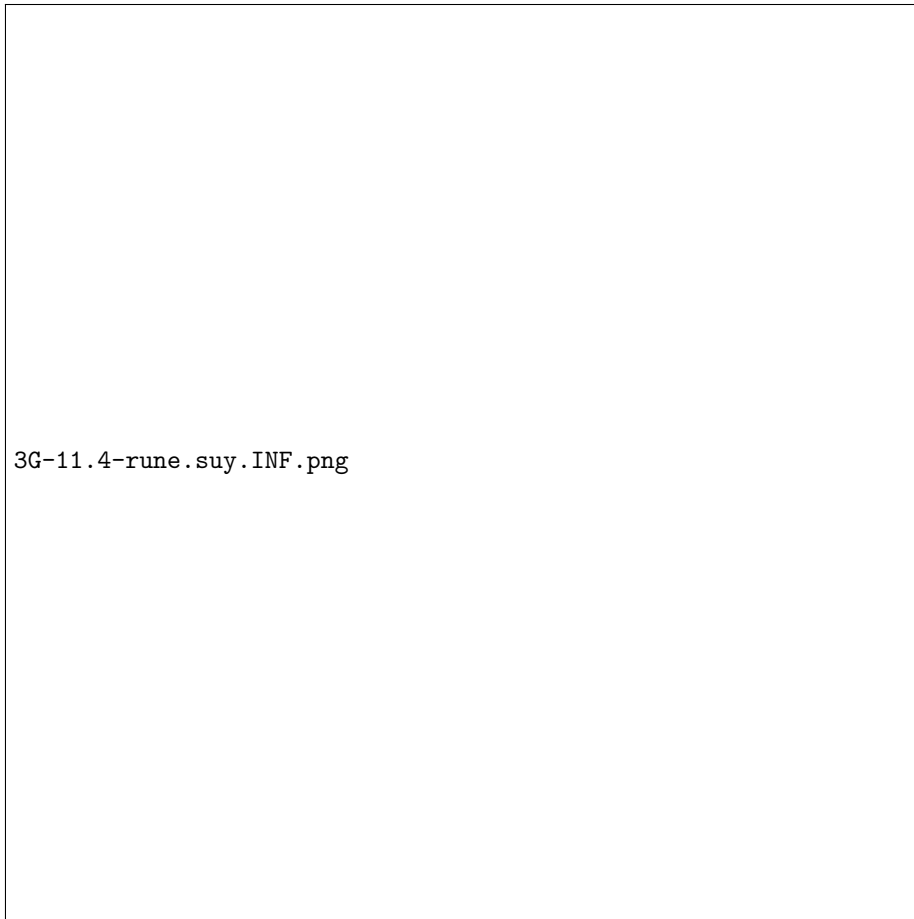
$$f''(x) = d\left(\frac{3x^2 - 12x + 15}{(x - 2)^2}\right) = \frac{(6x - 12)(x^2 - 4x + 4) - (2x - 4)(3x^2 - 12x + 15)}{(x - 2)^4}$$

$$= \frac{-6x + 12}{(x - 2)^4} = \frac{-6(x - 2)}{(x - 2)^4} = \frac{-6}{(x - 2)^3}$$

TO:

x	2		
T	-	-	-
N	-	0	+
$f''(x)$	+	⁽³⁾	-
$f(x)$	∪		∩

6. Schets:



References